

Homework 3 (1.8.11 ~ 1.10.3)

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1.8.11

(1) Equation of motion $\frac{d^2}{dt^2} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -\frac{2k}{m} & \frac{k}{m} \\ \frac{k}{m} & -\frac{2k}{m} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

Set $\begin{vmatrix} -\frac{2k}{m} + \omega^2 & \frac{k}{m} \\ \frac{k}{m} & -\frac{2k}{m} + \omega^2 \end{vmatrix} = 0$, we have $\omega_1 = \sqrt{\frac{3k}{m}}$, $\omega_2 = \sqrt{\frac{k}{m}}$.

The corresponding eigenvectors are $|\omega_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, $|\omega_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Set $\Lambda = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$, $\Lambda^\dagger \begin{pmatrix} -\frac{2k}{m} & \frac{k}{m} \\ \frac{k}{m} & -\frac{2k}{m} \end{pmatrix} \Lambda = \begin{pmatrix} -\omega_1^2 & 0 \\ 0 & -\omega_2^2 \end{pmatrix}$.

In eigenbasis,

$$\begin{pmatrix} \ddot{x}_I \\ \ddot{x}_{II} \end{pmatrix} = \begin{pmatrix} -\omega_1^2 & 0 \\ 0 & -\omega_2^2 \end{pmatrix} \begin{pmatrix} x_I \\ x_{II} \end{pmatrix}$$

$$\Rightarrow \begin{cases} x_I(t) = x_I(0) \cos \omega_1 t \\ x_{II}(t) = x_{II}(0) \cos \omega_2 t \end{cases} \quad (1)$$

In this problem, $|1\rangle = \begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

We first transform it into eigenbasis,

$$\begin{pmatrix} x_I(0) \\ x_{II}(0) \end{pmatrix} = \Lambda^\dagger \begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

In the eigenbasis, the state evolves according to (1),

$$\begin{pmatrix} x_I(t) \\ x_{II}(t) \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \cos \omega_1 t \\ \frac{1}{\sqrt{2}} \cos \omega_2 t \end{pmatrix}$$

Then, we transform it back to the original basis,

$$\begin{aligned} |1(t)\rangle &= \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = \Lambda \begin{pmatrix} x_I(t) \\ x_{II}(t) \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \cos \omega_1 t \\ \frac{1}{\sqrt{2}} \cos \omega_2 t \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \cos \omega_1 t + \frac{1}{2} \cos \omega_2 t \\ -\frac{1}{2} \cos \omega_1 t + \frac{1}{2} \cos \omega_2 t \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{2} \cos \sqrt{\frac{3k}{m}} t + \frac{1}{2} \cos \sqrt{\frac{k}{m}} t \\ -\frac{1}{2} \cos \sqrt{\frac{3k}{m}} t + \frac{1}{2} \cos \sqrt{\frac{k}{m}} t \end{pmatrix} \end{aligned}$$

(2) By substituting $\begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ in Eq. (1.8.9), we can get the same solution.

$$\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \cos \sqrt{\frac{k}{m}} t + \frac{1}{2} \cos \sqrt{\frac{3k}{m}} t \\ \frac{1}{2} \cos \sqrt{\frac{k}{m}} t - \frac{1}{2} \cos \sqrt{\frac{3k}{m}} t \end{pmatrix}$$

1.8.12

$$(1) \quad |\ddot{x}(t)\rangle = \Omega |x(t)\rangle$$

$$|x(t)\rangle = U(t) |x(0)\rangle$$

$$\Rightarrow \frac{d^2}{dt^2} U(t) |x(0)\rangle = \Omega U(t) |x(0)\rangle$$

$$\Rightarrow \left(\frac{d^2}{dt^2} - \Omega \right) U(t) |x(0)\rangle = 0$$

Since $|x(0)\rangle$ is arbitrary, we have

$$\frac{d^2}{dt^2} U(t) - \Omega U(t) = 0.$$

(2) From 1.8.11, we know the $\Lambda = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$ can diagonalize Ω , and therefore, it can also diagonalize U .

In this common basis, we have

$$\begin{pmatrix} \ddot{u}_{11}(t) & 0 \\ 0 & \ddot{u}_{22}(t) \end{pmatrix} - \begin{pmatrix} -\omega_1^2 & 0 \\ 0 & -\omega_2^2 \end{pmatrix} \begin{pmatrix} u_{11}(t) & 0 \\ 0 & u_{22}(t) \end{pmatrix} = 0$$

$$\Rightarrow \begin{cases} \ddot{u}_{11}(t) + \omega_1^2 u_{11}(t) = 0 \\ \ddot{u}_{22}(t) + \omega_2^2 u_{22}(t) = 0 \end{cases}$$

$$\Rightarrow \begin{cases} u_{11} = A_1 \cos \omega_1 t + B_1 \sin \omega_1 t \\ u_{22} = A_2 \cos \omega_2 t + B_2 \sin \omega_2 t \end{cases}$$

$$|\dot{x}(0)\rangle = \frac{d}{dt} [U(t) |x(0)\rangle] \Big|_{t=0} = \left(\frac{d}{dt} U(t) \right) \Big|_{t=0} |x(0)\rangle = 0$$

$$\Rightarrow \frac{d}{dt} U(t) \Big|_{t=0} = 0 \Rightarrow \dot{u}_{11}(0) = \dot{u}_{22}(0) = 0 \Rightarrow B_1 = B_2 = 0.$$

To satisfy that U is unitary, $A_1 = A_2 = 1$.

Therefore,

$$U = \begin{pmatrix} \cos \omega_1 t & 0 \\ 0 & \cos \omega_2 t \end{pmatrix},$$

which is the same as Eq. (1.8.43).

1.9.1

In the eigenbasis, $\Omega = \begin{pmatrix} \omega_1 & & \\ & \omega_2 & \\ & & \ddots \\ & & & \omega_m \end{pmatrix}$, where ω_i are eigenvalues.

$$f(\Omega) = \sum_{n=0}^{\infty} \Omega^n = \begin{pmatrix} \sum_{n=0}^{\infty} \omega_1^n & & \\ & \sum_{n=0}^{\infty} \omega_2^n & \\ & & \ddots \\ & & & \sum_{n=0}^{\infty} \omega_m^n \end{pmatrix} = \begin{pmatrix} \frac{1}{1-\omega_1} & & \\ & \frac{1}{1-\omega_2} & \\ & & \ddots \\ & & & \frac{1}{1-\omega_m} \end{pmatrix} = \frac{1}{1-\Omega}.$$

This equality holds iff $|\omega_i| < 1$ for $i = 1, \dots, m$.

Therefore, $f(\Omega)$ can be defined as $\frac{1}{1-\Omega}$ if and only if the absolute value of each of Ω 's eigenvalue is less than 1.

1.9.2

$$U^\dagger = (e^{iH})^\dagger = e^{-iH^\dagger} = e^{-iH}$$

$\uparrow$ H is Hermitian

$$U^\dagger U = e^{-iH} e^{iH} = e^{-iH+iH} = 1 \leftarrow \text{Unitary}$$

$\uparrow$ $\star$ This equality only holds for commuting operators.

1.9.3

In the eigenbasis of H , $U = e^{iH} = \begin{pmatrix} \sum_{n=0}^{\infty} \frac{(i\epsilon_1)^n}{n!} & & \\ & \ddots & \\ & & \sum_{n=0}^{\infty} \frac{(i\epsilon_m)^n}{n!} \end{pmatrix} = \begin{pmatrix} e^{i\epsilon_1} & & \\ & \ddots & \\ & & e^{i\epsilon_m} \end{pmatrix}$

Where $\epsilon_1, \dots, \epsilon_m$ are eigenvalues of H . i.e. $H = \begin{pmatrix} \epsilon_1 & & \\ & \ddots & \\ & & \epsilon_m \end{pmatrix}$

Therefore,

$$\det U = \prod_{i=1}^m e^{i\epsilon_i} = e^{i \sum_{i=1}^m \epsilon_i} = e^{i \text{Tr} H}$$

1.10.1

Since $\delta(x) = \delta(-x)$, we have $\delta(ax) = \delta(|a|x)$.

Therefore,

$$\int_{-\infty}^{\infty} \delta(ax) dx = \int_{-\infty}^{\infty} \delta(|a|x) dx = \int_{-\infty}^{\infty} \delta(|a|x) \cdot \frac{1}{|a|} d(|a|x)$$

$$= \frac{1}{|a|} \int_{-\infty}^{\infty} \delta(|a|x) d(|a|x) \quad (\text{change } |a|x \text{ to } x)$$

$$= \frac{1}{|a|} \int_{-\infty}^{\infty} \delta(x) dx$$

$$\text{Thus, } \delta(ax) = \delta(x)/|a|.$$

1.10.2

Expand $f(x)$ around x_i , where $f(x_i) = 0$:

$$\begin{aligned}
 f(x) &= f(x_i) + f'(x_i)(x-x_i) + \frac{1}{2}f''(x_i)(x-x_i)^2 + \dots \\
 &= 0 + f'(x_i)(x-x_i) + o[(x-x_i)^2] \\
 &\approx f'(x_i)(x-x_i)
 \end{aligned}$$

Introduce a test function $g(x)$,

$$\begin{aligned}
 \int_{-\infty}^{\infty} g(x) \delta(f(x)) dx &= \sum_i \int_{x_i-\epsilon}^{x_i+\epsilon} g(x) \delta(f(x)) dx \\
 &= \sum_i \int_{x_i-\epsilon}^{x_i+\epsilon} g(x) \delta\left(\left.\frac{df}{dx}\right|_{x=x_i} (x-x_i)\right) dx \\
 &= \sum_i \int_{x_i-\epsilon}^{x_i+\epsilon} g(x) \delta\left[\left|\left.\frac{df}{dx}\right|_{x=x_i}\right| (x-x_i)\right] dx \\
 &\stackrel{\text{See 1.10.1}}{\longrightarrow} \sum_i \int_{x_i-\epsilon}^{x_i+\epsilon} g(x) \frac{\delta(x-x_i)}{\left|\left.\frac{df}{dx}\right|_{x=x_i}\right|} dx
 \end{aligned}$$

Therefore, $\delta(f(x)) = \sum_i \frac{\delta(x-x_i)}{\left|\left.\frac{df}{dx}\right|_{x=x_i}\right|}.$

1.10.3

Introduce a test function $g(x)$. * $g(\infty)$ and $g(-\infty)$ are finite.

$$\begin{aligned}
 &\int_{-\infty}^{\infty} g(x) \frac{d}{dx} \theta(x-x') dx \\
 &= \int_{-\infty}^{\infty} g(x) d\theta(x-x') \\
 &= \theta(x-x') g(x) \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \theta(x-x') g'(x) dx \\
 &= 1 \cdot g(\infty) - 0 \cdot g(-\infty) - \int_0^{\infty} g'(x) dx \\
 &= g(\infty) - [g(\infty) - g(0)] \\
 &= g(0) \\
 &= \int_{-\infty}^{\infty} g(x) \delta(x) dx.
 \end{aligned}$$

Therefore, $\frac{d}{dx} \theta(x-x') = \delta(x).$